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Research Paper:

Multimodal Facial Emotion Recognition Using Improved Convolution Neural Networks Model

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In the quest for human-robot interaction (HRI), leading to the development of emotion recognition, learning, and analysis capabilities, robotics plays a significant role in human perception, attention, decision-making, and social communication. However, the accurate recognition of emotions in HRI remains a challenge. This is due to the coexistence of multiple sources of information in utilizing multimodal facial expressions and head poses as multiple convolutional neural networks (CNN) and deep learning are combined. This research analyzes and improves the robustness of emotion recognition, and proposes a novel approach that optimizes traditional deep neural networks that fall into poor local optima when optimizing the weightings of the deep neural network using standard methods. The proposed approach adaptively finds the better weightings of the network, resulting in a hybrid genetic algorithm with stochastic gradient descent (HGASGD). This hybrid algorithm combines the inherent, implicit parallelism of the genetic algorithm with the better global optimization of stochastic gradient descent (SGD). An experiment shows the effectiveness of our proposed approach in providing complete emotion recognition through a combination of multimodal data, CNNs, and HGASGD, indicating that it represents a powerful tool in achieving interactions between humans and robotics. To validate and test the effectiveness of our proposed approach through experiments, the performance and reliability of our approach and two variants of HGASGD FER are compared using a large dataset of facial images. Our approach integrates multimodal information from facial expressions and head poses, enabling the system to recognize emotions better. The results show that CNN-HGASGD outperforms CNNs-SGD and other existing state-of-the-art methods in terms of FER.

Keywords: emotion recognition, facial expression, head pose, convolution neural network, stochastic gradient descent

1. Introduction

Humans can express their emotions either through speech or gestures. Expressions of human emotion through facial expressions are a critical part of everyday communication. As they express countless emotions without spoken words, facial expressions are vital tools for communicating nonverbal cues, playing an important role in interpersonal relations. Therefore, researchers have rapidly grown interest in understanding emotions based on human-robot interaction (HRI). Foroni and Semin [1] presented a few necessities for human interaction abilities that machines possess, namely, the capability to respond, perceive, and understand human behavior for smooth social interactions, mainly expressed through understanding expressions of emotion. As the demand and implementation of robotics in our day-to-day application increase, facial recognition is no longer enough for smooth communication without understanding human emotions [2]. Hence, understanding human emotions is a significant challenge, as intentions occasionally differ. Understanding emotional intention requires that proper datasets be obtained first, and with facial covering, most of the expression is crucial over other means of expressing emotions [3].

Moreover, essential information is driven by facial expression stimulus for HRI reflection and recognition, and this has recently received much attention [4]. Facial detection and recognition are the primary step in FER (FER). As captured and processed images do not consist of faces only, training in facial detection systems is crucial to detecting faces in images with complex backgrounds, although the faces would be easy for humans to analyze and predict [5]. Therefore, facial detection

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requires separating faces from their backgrounds in processed data, with trained systems needing to first detect faces as inputs without background images cluttering the data. Present advances in facial recognition techniques rely on either grayscale or color image inputs based on window- or pixel-based techniques [6].

Various nations have sought to advance the existing technology over the years in order to understand FER better, ensure its stability, and optimize it for smooth HRI. Given that the face alone expresses more than half of the existing human emotions for daily communication and application, it is the center of the actual focus needed for the recognition of emotions [7]. The application of facial expressions differs from that of words, as words have precise interpretations. Facial expressions have a variety of meanings, as human beings exhibit numerous facial expressions that are generally interpreted in varied ways [8]. Emotions focus on the primary central affective states [9, 10]: angry, happy, sad, surprised, disgusted, fearful, anxious, bored, puzzled, uncertain, and neutral. Consider the expression of "Honestly, I didn't take the last apple from the fridge." It goes beyond facial expressions when, rather than focusing on the expression when reading the face, one should focus on emotions [2, 11] associated with accurate configurations scientifically recorded for facial purposes [3, 12]. These emotions, known as primary emotions, are suitable for all facial expressions, indivisible to other multi-cultures across the globe [13]. It is essential to note that the approach to facial expression detection should not be mistaken for human emotion detection. However, since emotions activate facial muscles, facial expression detection may be able to detect human emotions.

FER has emerged as a prominent research area in affective computing. The rich information from the nonverbal communication channels of the human face surpasses other modalities, such as speech and gestures [14]. It is important to note that the seven primary emotions are not the only emotions conveyed by humans universally but rather the ones with widely recognized expressions with reasonable amounts of data. For HRI to achieve high accuracy in facial expressions of emotion recognition, it is essential that the ability to capture subtle and imperceptible movements that humans cannot see is developed [15]. To improve facial expression recognition accuracy, HRI must focus on recognizing the basic emotions by analyzing the entire face, as facial expressions can involve multiple facial areas. For example, the moving of the eyebrows and upper eyelids as well as the pulling down or dropping of the jaw, are fundamental actions commonly associated with surprise. However, a common mistake in facial expression recognition is to assume that an expression represents a particular emotion without seeing all the movements [16]. Thus, eyebrow-raising is a joint movement associated with various emotions, such as fear, and is also used to accentuate specific parts of speech. This highlights the need to examine the entire face. The stability and increased demand for facial expression recognition accuracy are contingent on focusing on changes across the face.

The specific contributions of this paper are the following.

a) We introduced the optimization of the networks using a hybrid genetic algorithm with stochastic gradient descent (HGASGD) that aims to minimize classification errors by optimizing the traditional convolutional neural network (CNN) weighting in deep learning (DL) for FER in HRI.

b) The optimized weights of stochastic gradient descent (SGD) are used as the initial population of genetic algorithm (GA) to speed up the convergence and optimization, leading to improved performance in DL.

c) The experiment to evaluate the proposed method's effectiveness showed promising results in achieving complete emotion recognition through a combination of multimodal data, CNNs, and HGASGD, which represent tools toward the main contribution.

d) A HGASGD was developed for an improved CNN-HGASGD and compared to the traditional CNNs-SGD, along with state-of-the-art models.

HGASGD minimizes classification error and optimizes CNN weighting in DL. A HGASGD is developed into improved CNN-HGASGD, showing the outperformance difference with the traditional CNNs-SGD models. By combining the classifiers generated by the GA, the weight of the CNN is optimized, leading to improved performance in DL. The optimized weights of SGD are used as the initial population of GA to speed up the convergence. The subsequent sections of this paper are organized as follows. Section 2 thoroughly reviews facial expression and head pose extraction. Section 3 presents the CNN-HGASGD and DL model for understanding emotions. Section 4 presents the results, analysis, and discussion of the experiment.

2. Related Works

It is essential to have reliable methods for recognizing emotions. In recent years, more research attempts have been carried out in automated facial expression analysis [17] due to its significance in various fields for advanced purposes, including affective computing, clinical psychology, neuroscience, education and teaching, and forensic science [18]. The HRI field has been increasingly interested in developing models supporting engagement and turn-taking. "Connection events" for a robot to maintain commitment with a human interaction partner have been identified [19, 20]. Other systems have been developed to control multimodal dialogs for social robots, and several methods have been developed to enable robots to maintain engagement during HRI. For instance, some systems have been designed to control multimodal dialogs by switching behaviors in speech and gesture modalities [21], and others have used parallelized

processes with interruption handling for interruption task-based dialogues [22].

Additionally, there have been efforts to create autonomous robot controllers that can effectively engage in turn-taking interactions that resemble human-like interactions [23, 24]. Another critical factor in human learning is the "like-me" bias [25], which involves an early propensity and ability to map between actions observed in others and one's own actions. This concept has been demonstrated in early childhood [26]. In this light, facial recognition technology (FERET) has been applied to the estimation of age based on facial asymmetry, and it has been concluded that comparing right-left face asymmetry helps the estimation of age [27]. To overcome heterogenous facial pose appearance issues, subject-specific descriptors utilizing a three-dimensional pose-invariant method have been employed [28]. Additionally, the development of facial expression detection [29] has advanced significantly and has resulted in improvements in disease detection [30] and cognitive science [31].

The facial action coding system (FACS), a framework developed by Ekman et al. [32], provides a way to quantify facial muscle movement activity through the use of numeric codes called action units (AUs). Each AU corresponds to a specific facial muscle AU [33], such as cheek raising or brow lowering, and there are 46 AUs used to create different facial expressions associated with specific emotions [34]. Analyzing facial expressions by breaking them down into their AU components is essential, and there are two approaches to AU recognition: occurrence-based and intensity-based estimations. FACS defines five levels of AU intensity from A to E, with higher intensity values indicating a more robust occurrence of the corresponding facial muscle action [35]. Estimating AU intensity provides rich and detailed information about the subject's expression, and combining AUs into basic emotion classes is an effective way to quantify facial emotions. Despite the complexity and challenges of AU recognition, the translation of facial expressions into recognized basic emotion classes continues to be an active area of research.

Moreover, Barros et al. designed multimodal networks to extract features from multiple data sources and combine them for more robust and accurate results [36]. They proposed a multichannel CNN (MCCNN) for recognizing multimodal emotional states by extracting hierarchical features from visual stimuli. The model uses DL to learn high-level expressive features from multimodal data and then fuses them with intentional information to understand emotions. The model can extract edge-like features in the first layers and can generate more complex representations in deeper ones. It can be seen that neural networks can fuse different data types. An MCCNN with a DL model for emotion understanding in combination with the information of facial expressions and gestures has been proposed. In the proposal, MCCNN is used to learn high-level emotional features from multimodal data; these features are fused with intention information for understanding emotions.

In other works, Wu et al. [37] proposed a novel framework that merges local motion with appearance features to model the temporal dynamics of facial expressions and gestures. The framework uses MHI-HOG and Image-HOG features with temporal normalization (TN) or bag of words to capture motion and appearance information. The outputs from each branch can then be combined using a fusion layer, such as concatenation or pooling, to form a final representation for classification or other tasks. The idea of multimodal neural networks is introduced to combine information from different modalities. The goal of the recognition of images of facial emotion expression is to develop a classification system for identifying facial expressions. One approach uses stochastic gradient algorithms with a hybrid GA [38].

Another framework for FER that utilizes background images of the head pose has been proposed [39]. It characterizes the relevance of different facial patches, and experiments conducted on databases have demonstrated the significance of the head pose framework. However, obtaining high-level expression features can be challenging. DL has emerged as a powerful tool for unsupervised feature learning, leading to significant advancements in this field [40]. In the 1990s, Lecun et al. developed CNN systems representing DL specifically for handwritten digit recognition [41]. Extracting features from single images through a sequence of convolutional layers is a crucial feature of CNN systems.

DL refers to neural networks with multiple layers to identify complex patterns and features in input data, such as in images and speech. Due to its high accuracy and ability to handle large amounts of data, there has been a surge in interest in DL in recent years. Researchers, including Hinton [42], have demonstrated the viability of deep neural networks (DNNs) for image recognition. The DNNs use deep belief networks and deep automatic encoders to perform simple recognition and dimensionality reduction tasks. As a result, numerous researchers have started to apply DL to FER. Feature learning and grin detection are done simultaneously using the CNNDL [43]. Learned features have a remarkable capacity for discrimination. In CNN, high-level semantic information is often extracted by iteratively stacking convolutional and subsampling layers. As can be seen, feature learning and classification may be successfully combined into one model using DL.

AU provides the simplification of associating a facial expression with a corresponding emotion. However, selecting the best facial feature extraction methods is crucial if accurate recognition results are to be obtained, since facial expressions can vary greatly. An alternative approach to AU is using facial points (FP) analysis, which involves identifying critical points on a face, such as the eyes, nose, mouth, and eyebrows.

3. CNN

The ability to retrieve detailed emotional information from neural network theory is crucial to understanding

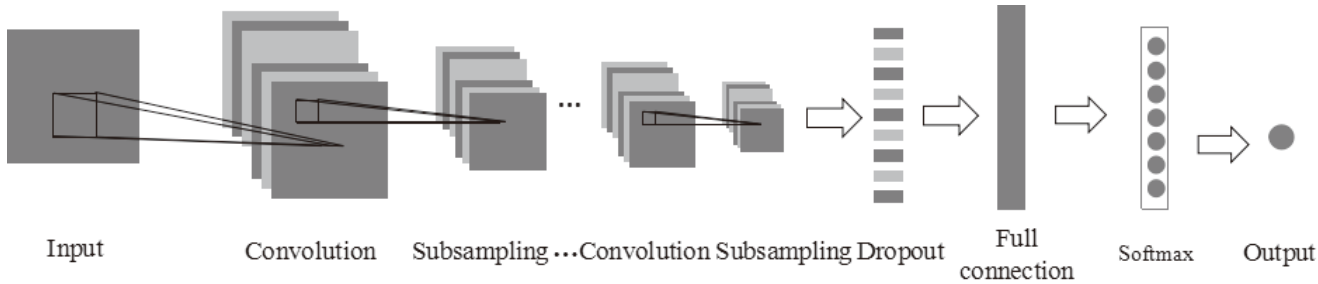

Fig. 1. CNN and DL model for understanding emotions.

emotional intentions. The retrieved data are one of the primary keys to promoting smooth interaction between humans and robots. To achieve the dynamic nature of such interactions, head poses and facial expressions are the key ways in which emotion is expressed. An emotional recognition algorithm based on a single modal communication interface is set to obtain accurate emotional information by providing the dynamic information necessary for subsequent emotional understanding. Different structural designs of CNNs are combined to extract high-level dynamic characteristics from speech and facial expression data. The model is based on canonical correlation analysis to fuse information from multiple models. This enables the extraction of discriminative features that capture multi-view emotional details, leading to the achievement of high accuracy and dynamic fusion capabilities and providing a powerful approach to emotion recognition.

3.1. Emotion Expression on CNN-HGASGD

To identify characteristics, CNN and DL combine high-level emotional features to extract additional data from various inputs. The system is constructed with multiple neural network structures that are based on the model-based canonical correlation analysis algorithm, pulling from low-level expressional features for high-level abstract emotional aspects, as illustrated in **Fig. 1**. The model also includes a softmax regression model that recognizes integrated expressive features and performs emotional recognition based on modal surface information.

The proposed method shows the optimizing flow of the proposed approach. We utilize n CNNs to extract a diverse range of features from n distinct input types. Each CNN comprises a complete connection layer followed by a softmax regression layer, resulting in a complete network. SGD is used to optimize the CNNs. The features generated by each CNN are merged in the feature fusion layer, with corresponding features from each network being combined. The resulting fusion features are input into the softmax layer for classification, forming the complete CNN-HGASGD. The CNNs consist of convolutional layers, subsampling layers that extract high-level semantic features, dropout layers that prevent overfitting, and a full connection layer. Finally, the CNN and DL model is optimized using the HGASGD method.

The convolutional layers use the convolution kernel to implement the local connections between neurons in the surrounding layer in extracting local association information from the PCA feature maps and high-level emotion data. The convolution process is carried out using the following.

$$x_j^{(l+1)} = f\left(\sum_{i=M_j} x_i^l \times W_{ij}^l + b_j\right), \quad (1)$$

where $x_j^{(l+1)}$ is the j-th feature map data of $(l+1)$-layer, $\sum_{i=M_j} x_i^l * W_{ij}^l$ is the convolution operation and summation for all feature map data of l-layer and the l-th layer of the j-th convolution kernel W, and b_k is the deviation corresponding to the k-th convolution kernel. The $f(\cdot)$ is a ReLu type activation function.

The convolution feature map generated by the convolutional layer is then input to the sub-sampling layer for average pooling, thereby weakening the sensitivity of the network to the change in the emotion feature position in the feature map. The average pooling implementation formula is as follows:

$$x_i^{(l+1)} = f\left(x_i^l \times \frac{1}{p^2}\right), \quad (2)$$

where x_i^l is the i-th convolution feature map in the l-layer, and p is the average pooling dimension.

The dropout layer can effectively solve the overfitting problem of the DNN and improve the efficiency of feature learning. In the training phase, it randomly does not allow all nodes of the input layer to work at each iteration, but their weights are preserved. The calculation process is shown in Eq. (3). In the test phase, all neurons of the network are activated to make the network into a complete structure, equivalent to the collection of multiple neural networks, the calculation process of which is shown in Eq. (4).

$$D_{Train}(x) = RandomZero(p) \times x, \quad (3)$$

$$D_{Test}(x) = (1-p) \times x. \quad (4)$$

Self-regularization (SR) layer is the output layer of the network to classify the learned emotion feature. For training set $\{(x^{(1)}, y^{(1)}), \ldots, (x^{(m)}, y^{(m)})\}$, there is $y^{(i)} \in \{1, 2, \ldots, k\}$. SR uses the cost function to evaluate its classification effect in the training phase. The definition of

softmax cost function is as follows:

$$J(\theta) = -\frac{1}{m}\left[\sum_{i=1}^{m}\sum_{j=1}^{k} 1\left\{y^{(i)} = j\right\}\log\frac{e^{\theta_j^T x^{(i)}}}{\sum_{l=1}^{k} e^{\theta_j^T x^{(i)}}}\right] + \frac{\lambda}{2}\sum_{i=1}^{k}\sum_{j=0}^{n}\theta_{ij}^2, \quad (5)$$

where $1\{y^{(i)} = j\}$ is an exponential function, and the value of the rule is $1\{\text{a true statement}\} = 1$ or $1\{\text{a false statement}\} = 0$.

The derivation of Eq. (5) is as follows:

$$\nabla_{\theta_j} J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[x^{(i)}\left(1\left\{y^{(i)} = j\right\} - p\left(y^{(i)} = j|x^{(i)};\theta\right)\right)\right] + \lambda\theta_j. \quad (6)$$

When the SR is being trained, Eq. (6) is substituted into the HGASGD to be minimized. The concrete implementation flow will be described in detail in Section 3.2.

3.2. A HGASGD for Optimization

The HGASGD is proposed to optimize the CNN for global search. HGASGD description of a GA is used in optimizing the weights of the CNNs in a CNN-HGASGD for emotional recognition.

The proposed method shows the optimizing flow of the proposed approach. The application of this work is given as follows.

a) Optimize the CNNs using SGD resulting in CNNs-SGD in optimizing the weightings.

b) Initialize a population of chromosomes randomly, one of which has the weightings obtained in step a).

c) Evaluate the fitness of each chromosome in the population by using a fitness function.

d) Select chromosomes for reproduction based on their fitness values, using stochastic universal sampling.

e) Apply genetic operations such as crossover and mutation to produce a new population. Add the chromosome with the best fitness value to the new population.

f) Repeat steps a)–e) until the maximum number of generations is reached and the best chromosome is obtained.

The weightings optimized by SGD will be taken as a chromosome of GA's initial population in the proposal. The remaining chromosomes are obtained by random initialization. SGD update is given by the following.

$$v_t = \gamma v_{t-1} + \alpha\nabla_\theta J\left(\theta; x^{(i)}, y^{(i)}\right), \quad (7)$$

$$\theta = \theta - v_t, \quad (8)$$

where $\nabla_\theta J(\theta; x^{(i)}, y^{(i)})$ is the gradient of the θ calculated from the training set $(x^{(i)}, y^{(i)})$, and α is a learning rate that prevents a large offset in a cost function. The v is the current velocity vector, which is the same as the dimension of θ. $\gamma \in (0,1]$ determines how many gradient iterations will be involved in the update.

The chromosome representations of GA are real-coded. At each genetic iteration, the fitness evaluation is needed first that is defined as Eq. (9), where $J(\theta)$ is defined as Eq. (6). Then, genetic operations of selection, crossover, and mutation are performed to produce a new population. However, the chromosomes with the best fitness will be directly passed on to the new population.

$$\text{fitness} = J(\theta). \quad (9)$$

Significantly, to improve the optimization efficiency of GA, the stochastic mini-batch is applied in the GA. This is inspired by the traditional gradient descent and SGD to reduce the use of computing resources. The data of the training set are to be randomly shuffled before each epoch. This stochastic mini-batch method is also used in SGD.

4. Experiment

The present research utilized a computing workstation with a 3.3 GHz Intel Core i5-4590 CPU processor and 6.0 GB of RAM. To test the feasibility of the proposed databases, an analysis benchmark database was employed. MATLAB R2018b was used as the analytical software.

4.1. Database Analysis

Our research used a sophisticated bimodal face and head pose (FABO) database [44] to conduct our study. The FABO database contains videos of 23 individuals from different nationalities exhibiting ten distinct emotional states: anger, anxiety, boredom, disgust, fear, happiness, puzzlement, sadness, surprise, and uncertainty. Each video is divided into four temporal phases: neutral, onset, apex, and offset. Only the frames from the apex phase for each emotional state were chosen for analysis, as this phase is considered the most significant for determining emotions. The FABO database provides an advanced and comprehensive dataset for analyzing facial expression and head pose data; such a dataset is critical to advancing research in affective computing and computer vision. Including multiple emotional states and subjects from diverse backgrounds ensures that the database is representative of real-world scenarios, making it a valuable resource for researchers in these fields.

Additionally, focusing on the apex phase allows for a more nuanced analysis of emotional states, providing researchers with a more detailed understanding of how emotions are expressed through facial expressions and head poses. The experiment involved selecting frames from the apex phase of each emotion and the neutral state, thereby

Table 1. Comparison of CNN-HGASGD with other state-of-the-art models.

Recognition index	Facial expression	Head pose	Fusion
CNN-HGASGD	82.14%	80.71%	91.43%
MCCNN [36]	70.70%	55.84%	90.30%
CNNs-SGD	79.29%	78.57%	88.57%
TN [37]	66.50%	66.70%	75.00%

increasing the number of emotional states to be classified from 10 to 11. A total of 649 facial expression frames and 649 head pose frames were then selected for further analysis. To prevent bias and ensure the accuracy of the results, a random sampling technique was used to split the data into 80% training and 20% test sets for both facial expressions and head poses. This approach facilitated a more comprehensive FABO database analysis. It enabled the development of a robust emotion recognition model for various applications.

Multimodal data can provide more comprehensive and accurate information about emotional states, and the proposed CNN-HGASGD model outperforms state-of-the-art models in terms of accuracy. This study contributes to emotion recognition by proposing a novel approach that utilizes both facial expressions and head poses for emotion recognition. This study also uses a new FABO database, which contains 10 emotional states performed by 23 subjects of different nationalities, providing a diverse set of emotional expressions for analysis. **Table 1** shows that the CNN-HGASGD outperformed the MCCNN, CNNs-SGD, and TN, with an average accuracy of 82.14% for facial expressions, 80.71% for head poses, and 91.43% for the multimodal data of facial expressions and head poses. **Fig. 2** shows the confusion matrix of CNN-HGASGD on multimodal facial expression and head pose data, where the accuracy of disgust, happiness, and uncertainty is 100%. The findings suggest that using multimodal data can improve emotion recognition accuracy, and the proposed CNN-HGASGD model can effectively classify emotions based on facial expressions and head poses.

4.2. Preparing Train and Test Image Sets

Our research aimed to analyze an extensive database of images and assess the accuracy of recognizing multiple facial expressions and head poses by setting out to demonstrate the potential of utilizing multiple approaches to accurately predict complex facial expressions, an approach which could have significant implications in various fields, including psychology and interaction. Our database analysis divided images into training, validation, and test sets. The training set contained approximately 1,388 images, the validation set 1,111, and the testing set 277. To avoid bias, we randomly divided the data into two batches and selected 80% of the images for training, with

	Angry	Anxiety	Boredom	Disgust	Fear	Happiness	Neutral	Puzzlement	Sadness	Surprise	Uncertainty
Uncertainty	0	0	0	0	0	0	0	0	0	0	**1**
Surprise	0	0	0	0	0	0	0	.25	0	**.75**	0
Sadness	0	0	0	.14	0	0	0	0	**.86**	0	0
Puzzlement	0	0	.13	0	0	0	0	**.87**	0	0	0
Neutral	0	0	0	0	0	0	**.96**	.04	0	0	0
Happiness	0	0	0	0	0	**1**	0	0	0	0	0
Fear	0	0	0	0	**.83**	.17	0	0	0	0	0
Disgust	0	0	0	**1**	0	0	0	0	0	0	0
Boredom	0	0	**.83**	0	0	0	0	.17	0	0	0
Anxiety	0	**.88**	0	.12	0	0	0	0	0	0	0
Angry	**.93**	0	0	0	0	0	.07	0	0	0	0

Fig. 2. Confusion matrix of CNN-HGASGD.

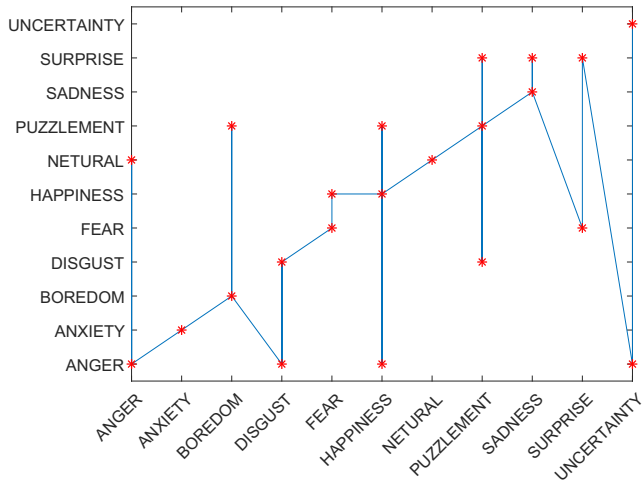

Fig. 3. Emotion viewpoints of CNN-HGASGD.

the remaining 20% left for testing. The ability to recognize blended multi-facial viewpoints is shown in **Fig. 3**. **Fig. 3** displays two of the eleven emotions used in the training. The interpretations of each feature boost the accuracy of multimodal facial expressions and head poses to an increased average of 86.43%, which correlates well with our findings.

4.3. Applied Layer Structure

The research structure aimed to analyze two types of modal data, facial expression data and head pose data, using CNNs. We utilized the Viola–Jones face detection algorithm to extract the facial area from facial expression frames [45]. We then normalized the facial expression and head pose frames to a uniform size 48×48 and 32×48, respectively. For the CNN of facial expression, we employed two convolutional layers and two average pooling layers with a convolution kernel size of 25×25 and 7×7, respectively. There was an average of two pooling dimensions.

Similarly, we used two convolutional layers and two intermediate pooling layers for the CNN of head poses, with a convolution kernel size of 15×25 and 6×7 and 40 and 50 filters, respectively. The p in the dropout layer was 0.5 in both CNNs. Our findings shed light on the specific parameters and architecture used to analyze the facial expression and head pose data using CNNs, which could benefit researchers in future works.

4.4. Results and Discussion

The proposed method involves configuring a human-robotic system and using it to acquire model data through HRI. This data are then transmitted to a computer workstation, where they are input into a trained CNN-HGASGD model to identify emotional states. The information about the emotional states is then sent back to the human-robotic system to achieve a more natural and harmonious interaction between a human and a robot. The experiments were conducted in a working environment where volunteers posed towards the system for passport headshots and were photographed while exhibiting various emotions. The data collected in this experiment were used to train and test the proposed method, which aims to improve the understanding of emotions and thereby the communication between humans and robots.

We trained both models in primary facial expression and head pose emotion recognition to quantify the suitability of CNNs-SGD and CNN-HGASGD models for characterizing multi-facial HRI. Similar parameters were utilized for both CNNs-SGD and CNN-HGASGD models by training and evaluating primary facial data and motion capture data to ascertain which model performed better in classifying emotions and expressions during HRI. Merging the datasets for facial expressions and related head poses improved yield results when combined and subjected to multiple iterations on a single CPU for approximately 80 s, resulting in significantly better output. The convergence line of fitness, as shown in **Figs. 4** and **5**, indicates that both the accuracy and loss improved over the course of the iterations. Regarding the reliability of corrections by SGDs, the outcomes for CNNs-SGD and CNN-HGASGD were almost indistinguishable in detecting facial emotions.

Table 1 shows that the CNN-HGASGD model performs better than CNNs-SGD and other existing methods in terms of FER. The multimodal fusion of CNN-HGASGD achieves an average accuracy of 91.43%, higher than that of different methods. The accuracy of CNN-HGASGD is 82.14% for facial expressions, 80.71% for head poses. To compare this with other methods, CNNs-SGD has an average accuracy of 79.29% and 78.57%, MCCNN [36] has an average accuracy of 70.70% and 55.84%, and TN [37] has an average accuracy of 66.50% and 66.70%, respectively, for facial expressions and head poses. These are promising results in emotion recognition. These techniques can potentially improve the accuracy and robustness of emotion recognition systems, which is a crucial factor in emotion recognition applications.

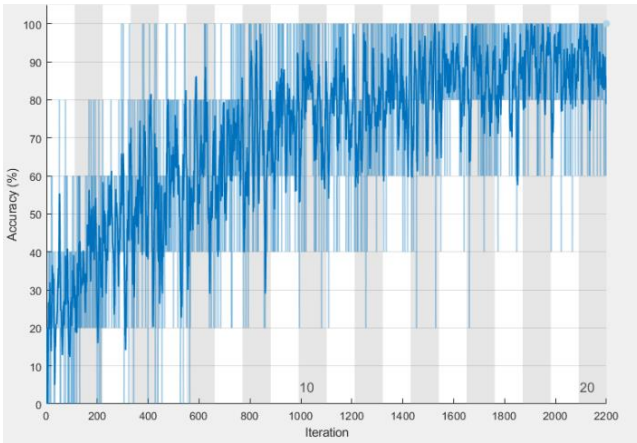

Fig. 4. Accuracy of different iterations.

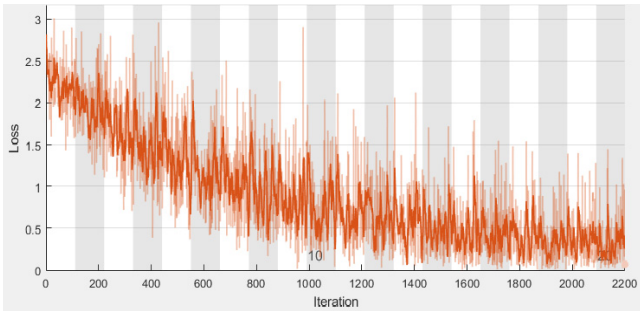

Fig. 5. Loss convergence line of fitness.

In terms of recognizing facial emotions, the CNN-HGASGD estimate outperformed CNNs-SGD. Overall, the reliability of CNNs-SGD and CNN-HGASGD for measuring FER is the same. The HRI system will receive feedback from the information on emotion understanding to develop a natural and harmonic interaction. Our prior research has shown that students have various behavioral goals depending on one current mode of feeling. As a result, the human-robotic system will decide on the volunteer's expression depending on their emotional state, gender, and preference for one or more other emotions. To validate the training and testing part of our experimental models, two of our research team volunteers helped out in our laboratory room. Experiments were carried out in various working-environment moods. These experiments mainly involved analyzing volunteers' numerous intentions as they worked on exhibiting random emotions and tried to communicate with the robots by striking different poses. Both posed and candid photos, including passport headshots and a few with rich background content, were also used.

The practical results of the confusion matrix from the volunteers' preliminary application are shown in **Fig. 6**. Demonstrating the applicability of our proposal, the average recognition accuracy on the application experiment was 82%. Out of the eleven primary central affective state frames, only seven states: angry, happy, neutral, sad, surprised, disgusted, and fearful were captured by our robot

	Angry	Disgust	Fear	Happiness	Neutral	Sadness	Surprise
Surprise	0	0	0	0	.42	0	**.58**
Sadness	0	0	0	0	0	**1**	0
Neutral	0	.42	0	0	**.58**	0	0
Happiness	0	0	0	**1**	0	0	0
Fear	0	0	**.58**	0	.42	0	0
Disgust	0	**1**	0	0	0	0	0
Angry	**1**	0	0	0	0	0	0

Fig. 6. Confusion matrix of the application experiment.

system as training sets from the volunteers. These frames also included modal data on facial expressions and head poses.

5. Conclusion

This paper combines the FABO dataset with the CNN-HGASGD-based model for reasonable accuracy in understanding and processing human emotions. The multi-facial modal expression is employed for emotional understanding by recognizing both the head pose with background content and the emotional states expressed on the face. The head pose and facial expressions emerge from lost features. The head pose fuses deep semantic detail features driven by the activations of the AU facial muscle features. The proposed method utilized in this paper displayed better prediction performance than the simple state-of-the-art and facial expression methods commonly used. The proposed method effectively optimizes the CNNs with the developed HGASGD to increase the efficiency and accuracy of predicting stable emotional states.

Based on the insight provided into the specific parameters and architecture utilized for analyzing multimodal expressions, future work will benefit and apply in different environments such as education and teaching. There are also plans to extend the experimental settings with background contents toward more efficient HRI by adding features geared towards effective models for identifying and analyzing the essential modalities for humans to communicate emotional intention and exploring new ideas and practical schemes for better advanced HRI.

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